

Trajectory_Analysis_PSORIASIS

June 1, 2026

```
[ ]: !pip install scanpy
      !pip install anndata
      !pip3 install igraph
      !pip install celltypist
      !pip install decoupler
      !pip install fa2-modified
      !pip install louvain
```

Collecting scanpy

Downloading scanpy-1.11.5-py3-none-any.whl.metadata (9.3 kB)

Collecting anndata>=0.8 (from scanpy)

Downloading anndata-0.12.3-py3-none-any.whl.metadata (9.9 kB)

Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.15.1)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.5.2)

Collecting legacy-api-wrap>=1.4.1 (from scanpy)

Downloading legacy_api_wrap-1.4.1-py3-none-any.whl.metadata (2.1 kB)

Requirement already satisfied: matplotlib>=3.7.5 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.10.0)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy) (8.4.0)

Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.5)

Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.60.0)

Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (2.0.2)

Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from scanpy) (25.0)

Requirement already satisfied: pandas>=1.5.3 in /usr/local/lib/python3.12/dist-packages (from scanpy) (2.2.2)

Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.0.2)

Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.13)

Requirement already satisfied: scikit-learn>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.6.1)

Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.16.2)

Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.13.2)

Collecting session-info2 (from scanpy)

 Downloading session_info2-0.2.3-py3-none-any.whl.metadata (3.4 kB)

Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.14.5)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy) (4.67.1)

Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)

Requirement already satisfied: umap-learn>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.9.post2)

Collecting array-api-compat>=1.7.1 (from anndata>=0.8->scanpy)

 Downloading array_api_compat-1.12.0-py3-none-any.whl.metadata (2.5 kB)

Collecting zarr!=3.0.*,>=2.18.7 (from anndata>=0.8->scanpy)

 Downloading zarr-3.1.3-py3-none-any.whl.metadata (10 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.3.3)

Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (2.9.0.post0)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba!=0.62.0rc1,>=0.57.1->scanpy) (0.43.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5.3->scanpy) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5.3->scanpy) (2025.2)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.3->scanpy) (3.6.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.7.5->scanpy) (1.17.0)

Collecting donfig>=0.8 (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)

 Downloading donfig-0.8.1.post1-py3-none-any.whl.metadata (5.0 kB)

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Collecting numcodecs>=0.14 (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
  Downloading numcodecs-0.16.3-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.3 kB)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (6.0.3)
Collecting crc32c>=2.7 (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
  Downloading crc32c-2.8-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (7.8
kB)
Downloading scanpy-1.11.5-py3-none-any.whl (2.1 MB)
2.1/2.1 MB
23.0 MB/s eta 0:00:00
Downloading anndata-0.12.3-py3-none-any.whl (169 kB)
169.3/169.3 kB
11.1 MB/s eta 0:00:00
Downloading legacy_api_wrap-1.4.1-py3-none-any.whl (10.0 kB)
Downloading session_info2-0.2.3-py3-none-any.whl (16 kB)
Downloading array_api_compat-1.12.0-py3-none-any.whl (58 kB)
58.2/58.2 kB
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Downloading zarr-3.1.3-py3-none-any.whl (276 kB)
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18.0 MB/s eta 0:00:00
Downloading donfig-0.8.1.post1-py3-none-any.whl (21 kB)
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numcodecs-0.16.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (8.8
MB)
8.8/8.8 MB
90.0 MB/s eta 0:00:00
Downloading crc32c-2.8-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (80 kB)
80.0/80.0 kB
4.8 MB/s eta 0:00:00
Installing collected packages: session-info2, numcodecs, legacy-api-wrap,
donfig, crc32c, array-api-compat, zarr, anndata, scanpy
Successfully installed anndata-0.12.3 array-api-compat-1.12.0 crc32c-2.8
donfig-0.8.1.post1 legacy-api-wrap-1.4.1 numcodecs-0.16.3 scanpy-1.11.5 session-
info2-0.2.3 zarr-3.1.3
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
packages (0.12.3)
Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from anndata) (1.12.0)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
packages (from anndata) (3.15.1)
Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata) (1.4.1)

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Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from anndata) (8.4.0)

Requirement already satisfied: numpy>=1.26 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.0.2)

Requirement already satisfied: packaging>=24.2 in /usr/local/lib/python3.12/dist-packages (from anndata) (25.0)

Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.2.2)

Requirement already satisfied: scipy>=1.12 in /usr/local/lib/python3.12/dist-packages (from anndata) (1.16.2)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata) (3.1.3)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata) (0.16.3)

Requirement already satisfied: typing-extensions>=4.9 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata) (4.15.0)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata) (6.0.3)

Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata) (2.8)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas!=2.1.2,>=2.1.0->anndata) (1.17.0)

Collecting igraph

 Downloading igraph-1.0.0-cp39-abi3-manylinux_2_28_x86_64.whl.metadata (4.4 kB)

Collecting texttable>=1.6.2 (from igraph)

 Downloading texttable-1.7.0-py2.py3-none-any.whl.metadata (9.8 kB)

Downloading igraph-1.0.0-cp39-abi3-manylinux_2_28_x86_64.whl (5.7 MB)

5.7/5.7 MB

43.8 MB/s eta 0:00:00

Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)

Installing collected packages: texttable, igraph

Successfully installed igraph-1.0.0 texttable-1.7.0

Collecting celltypist

 Downloading celltypist-1.7.1-py3-none-any.whl.metadata (45 kB)

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Requirement already satisfied: numpy>=1.19.0 in

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/usr/local/lib/python3.12/dist-packages (from celltypist) (2.0.2)
Requirement already satisfied: pandas>=1.0.5 in /usr/local/lib/python3.12/dist-
packages (from celltypist) (2.2.2)
Requirement already satisfied: scikit-learn>=0.24.1 in
/usr/local/lib/python3.12/dist-packages (from celltypist) (1.6.1)
Requirement already satisfied: openpyxl>=3.0.4 in
/usr/local/lib/python3.12/dist-packages (from celltypist) (3.1.5)
Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.12/dist-
packages (from celltypist) (8.3.0)
Requirement already satisfied: requests>=2.23.0 in
/usr/local/lib/python3.12/dist-packages (from celltypist) (2.32.4)
Requirement already satisfied: scanpy>=1.7.0 in /usr/local/lib/python3.12/dist-
packages (from celltypist) (1.11.5)
Collecting leidenalg>=0.9.0 (from celltypist)
  Downloading leidenalg-0.10.2-cp38-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (10 kB)
Collecting igraph<0.12,>=0.10.0 (from leidenalg>=0.9.0->celltypist)
  Downloading
igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(4.4 kB)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-
packages (from openpyxl>=3.0.4->celltypist) (2.0.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.0.5->celltypist) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.0.5->celltypist) (2025.2)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist)
(3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests>=2.23.0->celltypist) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist)
(2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist)
(2025.10.5)
Requirement already satisfied: anndata>=0.8 in /usr/local/lib/python3.12/dist-
packages (from scanpy>=1.7.0->celltypist) (0.12.3)
Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-
packages (from scanpy>=1.7.0->celltypist) (3.15.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages
(from scanpy>=1.7.0->celltypist) (1.5.2)
Requirement already satisfied: legacy-api-wrap>=1.4.1 in
/usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (1.4.1)

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Requirement already satisfied: matplotlib>=3.7.5 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
 (3.10.0)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
 packages (from scanpy>=1.7.0->celltypist) (8.4.0)

Requirement already satisfied: networkx>=2.7.1 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (3.5)

Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
 (0.60.0)

Requirement already satisfied: packaging>=21.3 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (25.0)

Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-
 packages (from scanpy>=1.7.0->celltypist) (1.0.2)

Requirement already satisfied: pynndescent>=0.5.13 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
 (0.5.13)

Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-
 packages (from scanpy>=1.7.0->celltypist) (1.16.2)

Requirement already satisfied: seaborn>=0.13.2 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
 (0.13.2)

Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-
 packages (from scanpy>=1.7.0->celltypist) (0.2.3)

Requirement already satisfied: statsmodels>=0.14.5 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
 (0.14.5)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
 (from scanpy>=1.7.0->celltypist) (4.67.1)

Requirement already satisfied: typing-extensions in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
 (4.15.0)

Requirement already satisfied: umap-learn>=0.5.6 in
 /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist)
 (0.5.9.post2)

Requirement already satisfied: threadpoolctl>=3.1.0 in
 /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.1->celltypist)
 (3.6.0)

Requirement already satisfied: array-api-compat>=1.7.1 in
 /usr/local/lib/python3.12/dist-packages (from
 anndata>=0.8->scanpy>=1.7.0->celltypist) (1.12.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
 /usr/local/lib/python3.12/dist-packages (from
 anndata>=0.8->scanpy>=1.7.0->celltypist) (3.1.3)

Requirement already satisfied: texttable>=1.6.2 in
 /usr/local/lib/python3.12/dist-packages (from
 igraph<0.12,>=0.10.0->leidenalg>=0.9.0->celltypist) (1.7.0)

Requirement already satisfied: contourpy>=1.0.1 in

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/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.3.3)
Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from
matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (3.2.5)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from
numba!=0.62.0rc1,>=0.57.1->scanpy>=1.7.0->celltypist) (0.43.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->pandas>=1.0.5->celltypist) (1.17.0)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-
packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist)
(0.8.1.post1)
Requirement already satisfied: numcodecs>=0.14 in
/usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14-
>zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.16.3)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from
donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist)
(6.0.3)
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-
packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8-
>scanpy>=1.7.0->celltypist) (2.8)
Downloading celltypist-1.7.1-py3-none-any.whl (7.3 MB)
7.3/7.3 MB
62.0 MB/s eta 0:00:00
Downloading
leidenalg-0.10.2-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.0
MB)
2.0/2.0 MB
54.4 MB/s eta 0:00:00
Downloading
igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.4 MB)
4.4/4.4 MB
86.0 MB/s eta 0:00:00
Installing collected packages: igraph, leidenalg, celltypist
Attempting uninstall: igraph
Found existing installation: igraph 1.0.0

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Uninstalling igraph-1.0.0:
  Successfully uninstalled igraph-1.0.0
Successfully installed celltypist-1.7.1 igraph-0.11.9 leidenalg-0.10.2
Collecting decoupler
  Downloading decoupler-2.1.1-py3-none-any.whl.metadata (8.6 kB)
Collecting adjusttext (from decoupler)
  Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-packages (from decoupler) (0.12.3)
Collecting docrep (from decoupler)
  Downloading docrep-0.3.2.tar.gz (33 kB)
  Preparing metadata (setup.py) ... done
Collecting marsilea (from decoupler)
  Downloading marsilea-0.5.6-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages (from decoupler) (0.60.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from decoupler) (2.32.4)
Collecting scipy<1.16 (from decoupler)
  Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(61 kB)

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62.0/62.0 kB

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Requirement already satisfied: session-info2 in
/usr/local/lib/python3.12/dist-packages (from decoupler) (0.2.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from decoupler) (4.67.1)
Requirement already satisfied: numpy<2.5,>=1.23.5 in
/usr/local/lib/python3.12/dist-packages (from scipy<1.16->decoupler) (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from adjusttext->decoupler) (3.10.0)
Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.12.0)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (3.15.1)
Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.4.1)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (8.4.0)
Requirement already satisfied: packaging>=24.2 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (25.0)
Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (2.2.2)
Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.1.3)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from docrep->decoupler) (1.17.0)

```


Collecting legendkit (from marsilea->decoupler)
 Downloading legendkit-0.3.6-py3-none-any.whl.metadata (2.6 kB)
 Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-packages (from marsilea->decoupler) (4.5.0)
 Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (from marsilea->decoupler) (0.13.2)
 Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba->decoupler) (0.43.0)
 Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.4.4)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.11)
 Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2.5.0)
 Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2025.10.5)
 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (1.3.3)
 Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (4.60.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (1.4.9)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (11.3.0)
 Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (3.2.5)
 Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (2.9.0.post0)
 Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)
 Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.8.1.post1)
 Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.16.3)
 Requirement already satisfied: typing-extensions>=4.9 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (4.15.0)
 Requirement already satisfied: pyarrow>=10.0.1 in

```

/usr/local/lib/python3.12/dist-packages (from
pandas[parquet]->marsilea->decoupler) (18.1.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-
packages (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (2.8)
Downloading decoupler-2.1.1-py3-none-any.whl (117 kB)
118.0/118.0 kB
7.7 MB/s eta 0:00:00
Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3
MB)
37.3/37.3 MB
24.7 MB/s eta 0:00:00
Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)
Downloading marsilea-0.5.6-py3-none-any.whl (84 kB)
85.0/85.0 kB
4.7 MB/s eta 0:00:00
Downloading legendkit-0.3.6-py3-none-any.whl (24 kB)
Building wheels for collected packages: docrep
  Building wheel for docrep (setup.py) ... done
  Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19876
sha256=11e61a8a03c04ec4a8a3a1c1f8b5d1116b91fa1586238c589ec3f434fe2900d7
  Stored in directory: /root/.cache/pip/wheels/d6/19/ee/0a6a1793d91c449563b49cca
b57ce52da3e6fab7614836bd8c
Successfully built docrep
Installing collected packages: scipy, docrep, legendkit, adjustttext, marsilea,
decoupler
  Attempting uninstall: scipy
    Found existing installation: scipy 1.16.2
    Uninstalling scipy-1.16.2:
      Successfully uninstalled scipy-1.16.2
Successfully installed adjustttext-1.3.0 decoupler-2.1.1 docrep-0.3.2
legendkit-0.3.6 marsilea-0.5.6 scipy-1.15.3
Collecting fa2-modified
  Downloading fa2_modified-0.3.10.tar.gz (474 kB)
474.3/474.3
kB 8.4 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (1.15.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages

```

```

(from fa2-modified) (4.67.1)
Building wheels for collected packages: fa2-modified
  Building wheel for fa2-modified (pyproject.toml) ... done
  Created wheel for fa2-modified:
filename=fa2_modified-0.3.10-cp312-cp312-linux_x86_64.whl size=493571
sha256=4375084166ac3258be2d6817f154927259c62a9de0117509456ae8c165335291
  Stored in directory: /root/.cache/pip/wheels/e7/b5/c2/e87afecc23dc78723045cde9
a733677ee2b3c1374a7ed3a5e8
Successfully built fa2-modified
Installing collected packages: fa2-modified
Successfully installed fa2-modified-0.3.10
Collecting louvain
  Downloading louvain-0.8.2.tar.gz (4.2 MB)
                                4.2/4.2 MB
34.5 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: igraph<0.12,>=0.10.0 in
/usr/local/lib/python3.12/dist-packages (from louvain) (0.11.9)
Requirement already satisfied: texttable>=1.6.2 in
/usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->louvain)
(1.7.0)
Building wheels for collected packages: louvain
  Building wheel for louvain (setup.py) ... done
  Created wheel for louvain: filename=louvain-0.8.2-cp312-cp312-linux_x86_64.whl
size=971836
sha256=4888ed3d50ee83f200b485ba2564d503db302f0cca73fef8391c5fe0b95e59e8
  Stored in directory: /root/.cache/pip/wheels/40/de/2b/bb7ed19d84727f9f299f20cd
34c42bba9c8bef7d83d2255c86
Successfully built louvain
Installing collected packages: louvain
Successfully installed louvain-0.8.2

```

```
[ ]: #Import core single cell datasets
```

```
import scanpy as sc
import anndata as ad
import numpy as np
```

```
[ ]: !wget https://ftp.ncbi.nlm.nih.gov/geo/series/GSE205nnn/GSE205790/suppl/
↪GSE205790_RAW.tar
```

```

--2025-10-24 04:08:09-- https://ftp.ncbi.nlm.nih.gov/geo/series/GSE205nnn/GSE20
5790/suppl/GSE205790_RAW.tar
Resolving ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)... 130.14.250.13,
130.14.250.31, 130.14.250.7, ...
Connecting to ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)|130.14.250.13|:443...
connected.
HTTP request sent, awaiting response... 200 OK

```

Length: 98334720 (94M) [application/x-tar]
Saving to: 'GSE205790_RAW.tar'

GSE205790_RAW.tar 100%[=====>] 93.78M 25.6MB/s in 3.9s

2025-10-24 04:08:13 (23.9 MB/s) - 'GSE205790_RAW.tar' saved [98334720/98334720]

```
[ ]: !tar -xvf GSE205790_RAW.tar
```

```
GSM6229474_C57IMQD_barcodes.tsv.gz
GSM6229474_C57IMQD_features.tsv.gz
GSM6229474_C57IMQD_matrix.mtx.gz
GSM6229475_C57IMQE_barcodes.tsv.gz
GSM6229475_C57IMQE_features.tsv.gz
GSM6229475_C57IMQE_matrix.mtx.gz
```

```
[ ]: !mkdir -p GSM6229474
!mkdir -p GSM6229475
!mv GSM6229474_* GSM6229474
!mv GSM6229475_* GSM6229475
```

```
[ ]: !pwd
```

/content

```
[ ]: !mv /content/GSM6229474/GSM6229474_C57IMQD_barcodes.tsv.gz /content/GSM6229474/
↪ barcodes.tsv.gz
!mv /content/GSM6229474/GSM6229474_C57IMQD_features.tsv.gz /content/GSM6229474/
↪ features.tsv.gz
!mv /content/GSM6229474/GSM6229474_C57IMQD_matrix.mtx.gz /content/GSM6229474/
↪ matrix.mtx.gz
```

```
!mv /content/GSM6229475/GSM6229475_C57IMQE_barcodes.tsv.gz /content/GSM6229475/
↪ barcodes.tsv.gz
!mv /content/GSM6229475/GSM6229475_C57IMQE_features.tsv.gz /content/GSM6229475/
↪ features.tsv.gz
!mv /content/GSM6229475/GSM6229475_C57IMQE_matrix.mtx.gz /content/GSM6229475/
↪ matrix.mtx.gz
```

```
[ ]: psorasis_mtx = sc.read_10x_mtx('GSM6229474/')
psorasis_mtx.var_names_make_unique()
```

```
[ ]: #psorasis_mtx = sc.read_10x_mtx('GSM6229475/')
#psorasis_mtx.var_names_make_unique()
```

```
[ ]: psorasis_mtx.shape
```

```
[ ]: (5376, 31053)
```

```
[ ]: psorasis_mtx.var.head()
```

```
[ ]:
      gene_ids      feature_types
Xkr4      ENSMUSG000000051951  Gene Expression
Gm1992     ENSMUSG000000089699  Gene Expression
Gm37381    ENSMUSG000000102343  Gene Expression
Rp1        ENSMUSG000000025900  Gene Expression
Sox17      ENSMUSG000000025902  Gene Expression
```

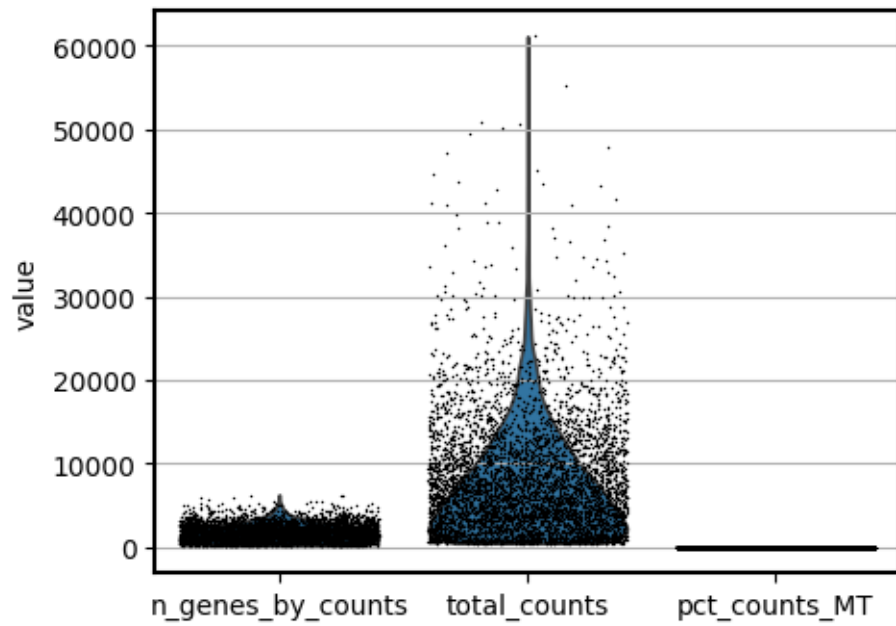
```
[ ]: psorasis_mtx.var['MT'] = psorasis_mtx.var_names.str.startswith("MT-")
psorasis_mtx.var['RIBO'] = psorasis_mtx.var_names.str.startswith("RPS", "RPL")
psorasis_mtx.var['HB'] = psorasis_mtx.var_names.str.startswith("^HB[^(P)]")
```

```
[ ]: sc.pp.calculate_qc_metrics(
      psorasis_mtx, qc_vars=["MT", 'RIBO', 'HB'], inplace=True, log1p=True
)
```

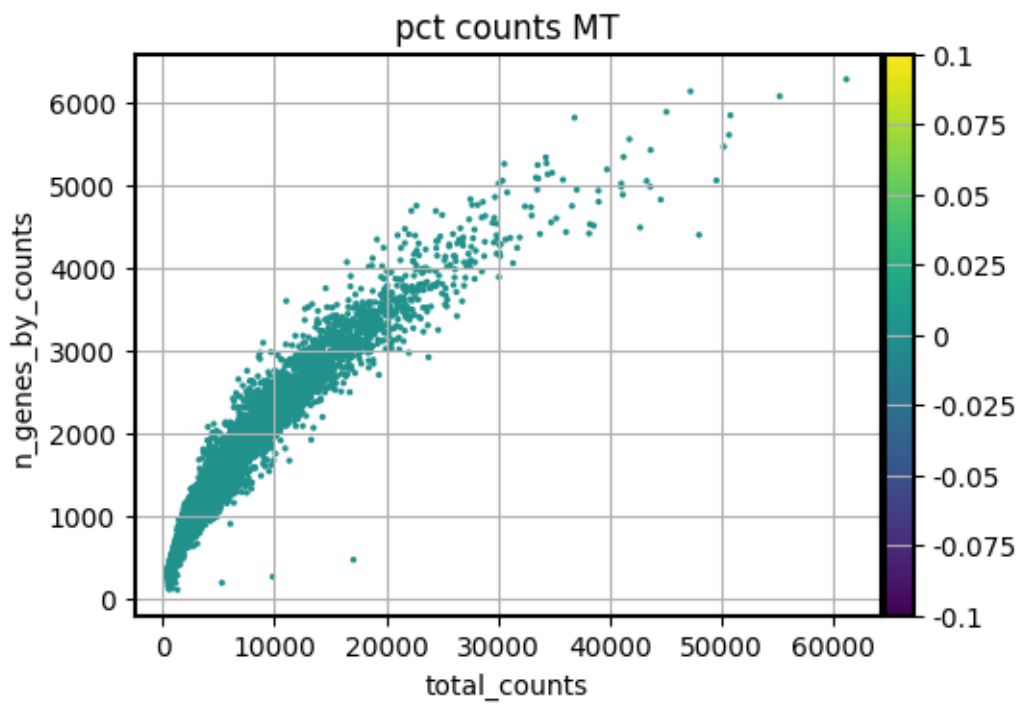
```
[ ]: import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (5,4) # Adjust figure size
plt.rcParams["axes.grid"] = True # Add grid to plots
plt.rcParams["axes.edgecolor"] = "black" # Set plot border color
plt.rcParams["axes.linewidth"] = 1.5 # Set plot border width
plt.rcParams["axes.facecolor"] = "white" # Set background color
plt.rcParams["axes.labelcolor"] = "black" # Set label color
plt.rcParams["xtick.color"] = "black" # Set x-axis tick color
plt.rcParams["ytick.color"] = "black" # Set y-axis tick color
plt.rcParams["text.color"] = "black" # Set text color
```

```
[ ]: sc.pl.violin(
      psorasis_mtx,
      ["n_genes_by_counts", 'total_counts', 'pct_counts_MT'],
      jitter=0.4,
      multi_panel=False,
)
```



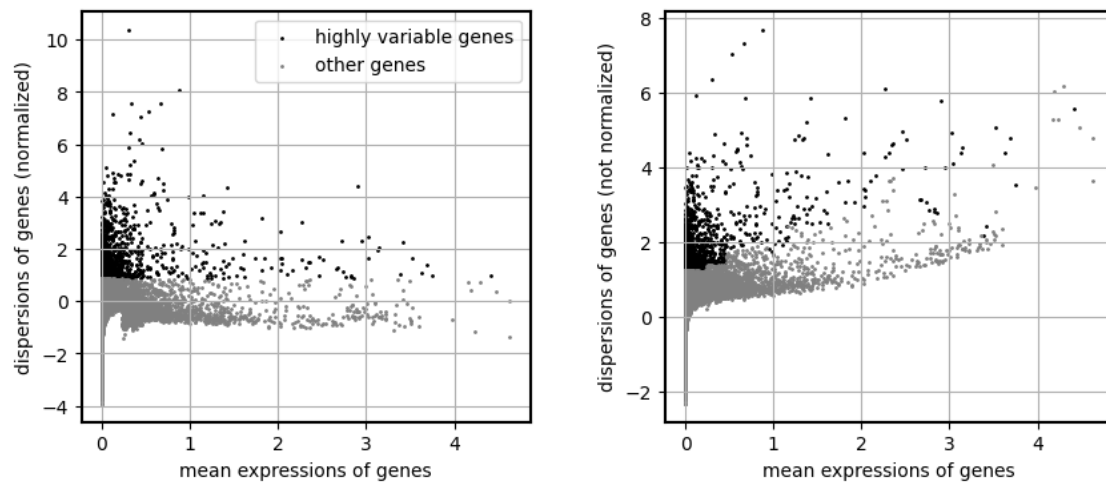
```
[ ]: sc.pl.scatter(psorasis_mtx, "total_counts", "n_genes_by_counts",
    ↪color="pct_counts_MT")
```



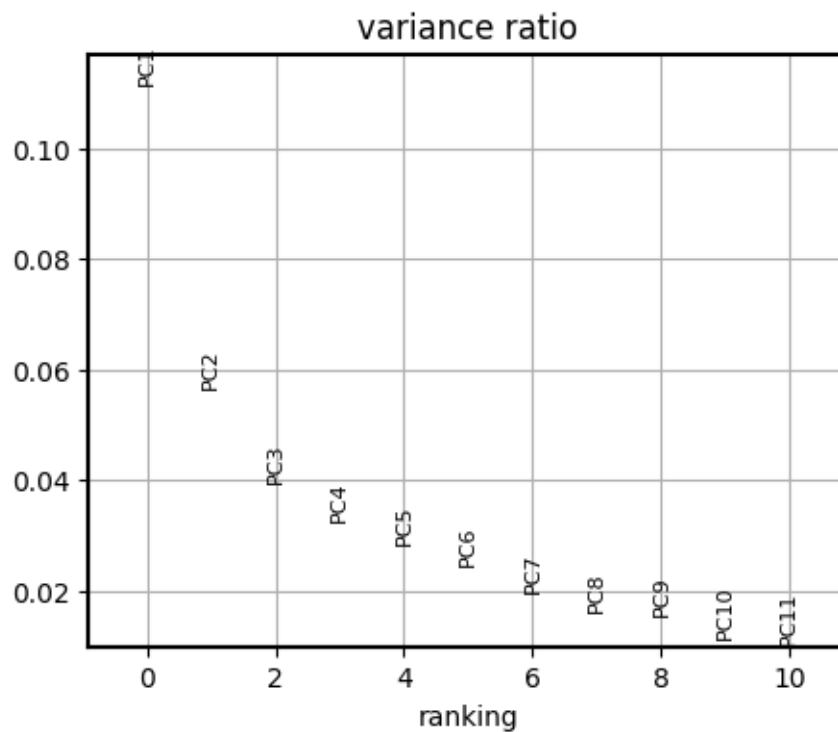
```
[ ]: sc.pp.scrublet(psorasis_mtx)
```

```
[ ]: #Normalisation
psorasis_mtx.layers["counts"] = psorasis_mtx.X.copy()
sc.pp.normalize_total(psorasis_mtx)
sc.pp.log1p(psorasis_mtx)
```

```
[ ]: #Feature selection
sc.pp.highly_variable_genes(psorasis_mtx, n_top_genes=2500)
sc.pl.highly_variable_genes(psorasis_mtx)
```



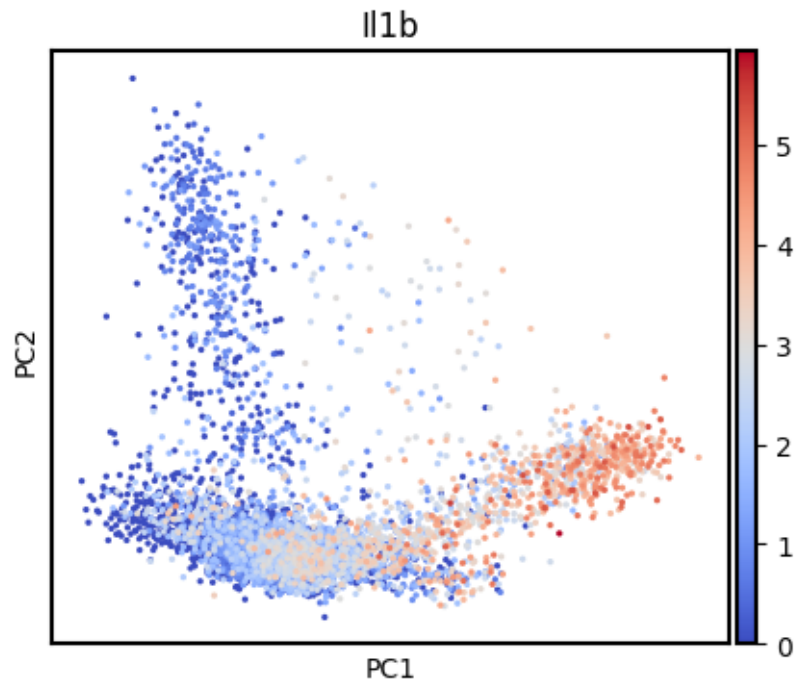
```
[ ]: #Dim Reduction
sc.tl.pca(psorasis_mtx)
sc.pl.pca_variance_ratio(psorasis_mtx, n_pcs=10, log=False)
```



```
[ ]: il_genes = [gene for gene in psoriasis_mtx.var_names[psoriasis_mtx.
    ↪var['highly_variable']] if gene.lower().startswith('il')]
print(il_genes)
```

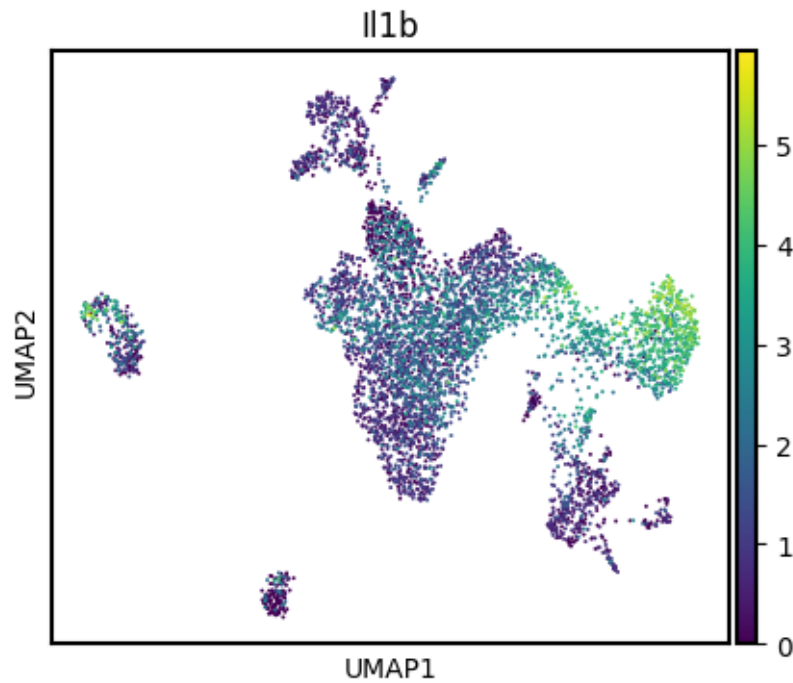
```
['Il17a', 'Il17f', 'Il1r2', 'Il1r1', 'Il19', 'Il10', 'Il2ra', 'Il1f9', 'Il1rn',
'I11a', 'I11b', 'I16', 'I15ra', 'I14i1', 'I122', 'I123a', 'I112b', 'I14',
'I113', 'I13', 'I17r', 'I12rb', 'Ildr1']
```

```
[ ]: sc.pl.pca(psoriasis_mtx, color="I11b", cmap="coolwarm")
```

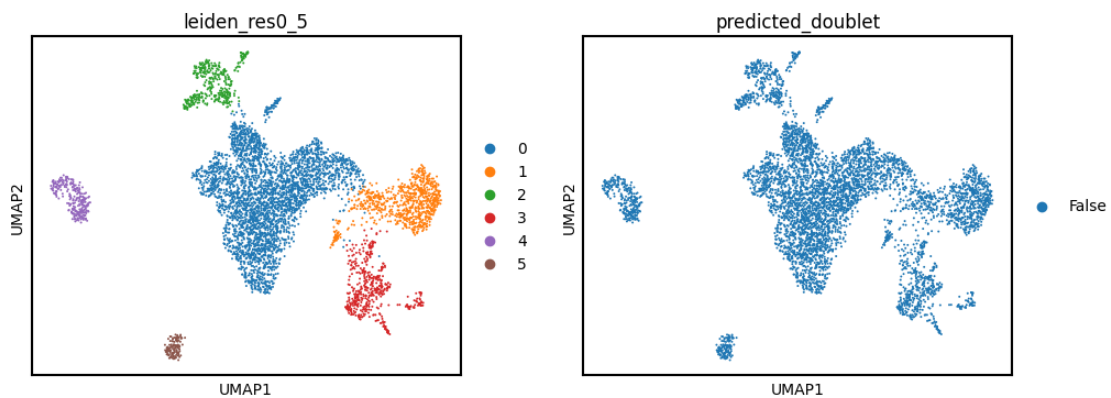
```
[ ]: sc.pp.neighbors(psorasis_mtx)
      sc.tl.umap(psorasis_mtx)
```

```
[ ]: sc.pl.umap(
      psorasis_mtx,
      color=["l1b"],
      size=8,
      )
```



```
[ ]: sc.tl.leiden(psorasis_mtx, flavor="igraph", n_iterations=2,
↳key_added="leiden_res0_5", resolution=0.05)
```

```
[ ]: sc.pl.umap(
    psorasis_mtx,
    color=["leiden_res0_5", 'predicted_doublet'],
    size=8,
)
```



```
[ ]: import decoupler as dc
```

```
[ ]: # Query Omnipath and get PanglaoDB
markers = dc.op.resource(name="PanglaoDB", organism="mouse")

# Keep canonical cell type markers alone
markers = markers[markers["mouse"]]

# Remove duplicated entries
markers = markers[~markers.duplicated(["cell_type", "genesymbol"])]

# Format because dc only accepts cell_type and genesymbol

markers = markers.rename(columns={"cell_type": "source", "genesymbol": "target"})
markers = markers[["source", "target"]]

markers.head()
```

```
[ ]:      source  target
0   Hepatocytes   A1cf
1  Bergmann glia   A2m
2  Bergmann glia   Mug2
3  Bergmann glia   Mug1
4    Mast cells  A4galt
```

```
[ ]: psoriasis_mtx.var_names
```

```
[ ]: Index(['Xkr4', 'Gm1992', 'Gm37381', 'Rp1', 'Sox17', 'Gm37323', 'Mrpl15',
          'Lypla1', 'Gm37988', 'Tcea1',
          ...,
          'AC125149.1', 'AC125149.2', 'AC125149.4', 'AC234645.1', 'AC168977.2',
          'AC168977.1', 'AC149090.1', 'CAAA01118383.1', 'Vmn2r122',
          'CAAA01147332.1'],
          dtype='object', length=31053)
```

```
[ ]: dc.mt.ulm(data=psoriasis_mtx,
              net=markers,
              tmin = 3)
```

```
[ ]: score = dc.pp.get_obs(mtx=psoriasis_mtx, key="score_ulm")
```

```
[ ]: psoriasis_mtx.obs[mtx="score_ulm"].head(1)
```

```
[ ]:      Acinar cells  Adipocyte progenitor cells  Adipocytes \
AAACCTGAGAGGTTAT-1      -1.261035                -0.326741   -0.927165

      Adrenergic neurons  Airway epithelial cells \
```

AAACCTGAGAGGTTAT-1	-0.687257	0.076385
	Airway goblet cells	Airway smooth muscle cells \
AAACCTGAGAGGTTAT-1	-0.53847	-0.532316
	Alpha cells	Alveolar macrophages \
AAACCTGAGAGGTTAT-1	1.32622	2.79075
	Anterior pituitary gland cells ...	Thymocytes \
AAACCTGAGAGGTTAT-1	-0.659633 ...	1.182331
	Transient cells	Trichocytes Trigeminal neurons \
AAACCTGAGAGGTTAT-1	0.099665	-0.752875 -1.188909
	Trophoblast cells	Trophoblast progenitor cells \
AAACCTGAGAGGTTAT-1	-0.606851	-0.532316
	Tuft cells	Undefined placental cells Urothelial cells \
AAACCTGAGAGGTTAT-1	1.734544	-1.019547 -0.813221
	Vascular smooth muscle cells	
AAACCTGAGAGGTTAT-1	-0.752875	

[1 rows x 173 columns]

```
[ ]: psoriasis_mtx.obsm["score_ulm"].columns
```

```
[ ]: Index(['Acinar cells', 'Adipocyte progenitor cells', 'Adipocytes',
          'Adrenergic neurons', 'Airway epithelial cells', 'Airway goblet cells',
          'Airway smooth muscle cells', 'Alpha cells', 'Alveolar macrophages',
          'Anterior pituitary gland cells',
          ...,
          'Thymocytes', 'Transient cells', 'Trichocytes', 'Trigeminal neurons',
          'Trophoblast cells', 'Trophoblast progenitor cells', 'Tuft cells',
          'Undefined placental cells', 'Urothelial cells',
          'Vascular smooth muscle cells'],
          dtype='object', length=173)
```

```
[ ]: #rank genes
psoriasis_gene_rank = dc.tl.rankby_group(score, groupby="leiden_res0_5",
    reference="rest", method="t-test_overestim_var")
psoriasis_gene_rank = psoriasis_gene_rank[psoriasis_gene_rank["stat"] > 0]
psoriasis_gene_rank.head(5)
```

	group	reference	name	stat	meanchange	\
21	0	rest	Dendritic cells	32.798415	1.733199	
28	0	rest	Proximal tubule cells	30.153956	0.534434	

53	0	rest	Delta cells	23.265208	0.288656
62	0	rest	Red pulp macrophages	21.392348	0.498373
77	0	rest	Plasma cells	19.085622	0.427474

	pval	padj
21	2.913403e-216	2.290994e-215
28	3.369830e-187	2.010278e-186
53	4.040271e-115	1.294383e-114
62	4.032925e-98	1.107454e-97
77	4.287698e-79	9.509895e-79

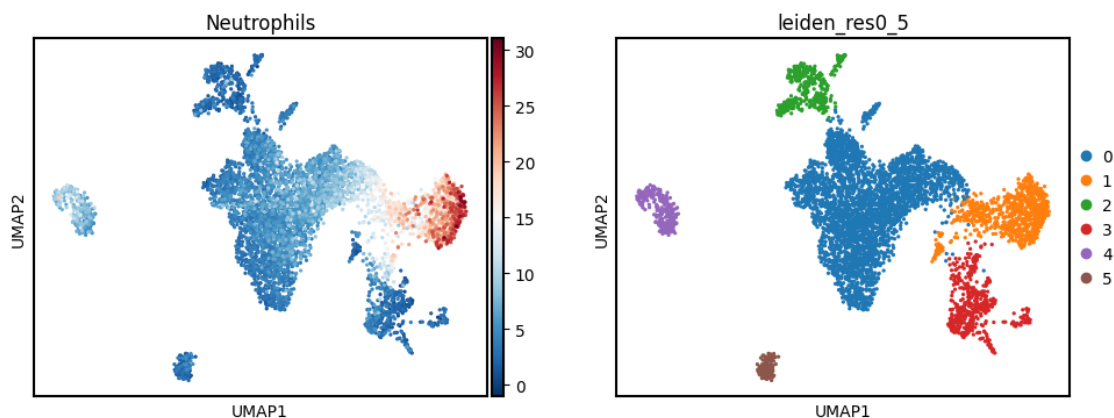
```
[ ]: top_cell_type_per_group = psoriasis_gene_rank.groupby('group')['name'].
      ↪apply(lambda x: x.head(1))
      display(top_cell_type_per_group.to_dict())
```

/tmp/ipython-input-552041747.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
top_cell_type_per_group =
psoriasis_gene_rank.groupby('group')['name'].apply(lambda x: x.head(1))
```

```
{('0', 21): 'Dendritic cells',
 ('1', 346): 'Neutrophils',
 ('2', 173): 'T memory cells',
 ('3', 519): 'Hepatic stellate cells',
 ('4', 692): 'Basophils',
 ('5', 865): 'Mast cells'}
```

```
[ ]: sc.pl.umap(score, color=["Neutrophils","leiden_res0_5"], cmap="RdBu_r")
```



```
[ ]: dict_ann = psoriasis_gene_rank[psoriasis_gene_rank["stat"] > 0].groupby("group").
      ↪head(1).set_index("group")["name"].to_dict()
dict_ann
```

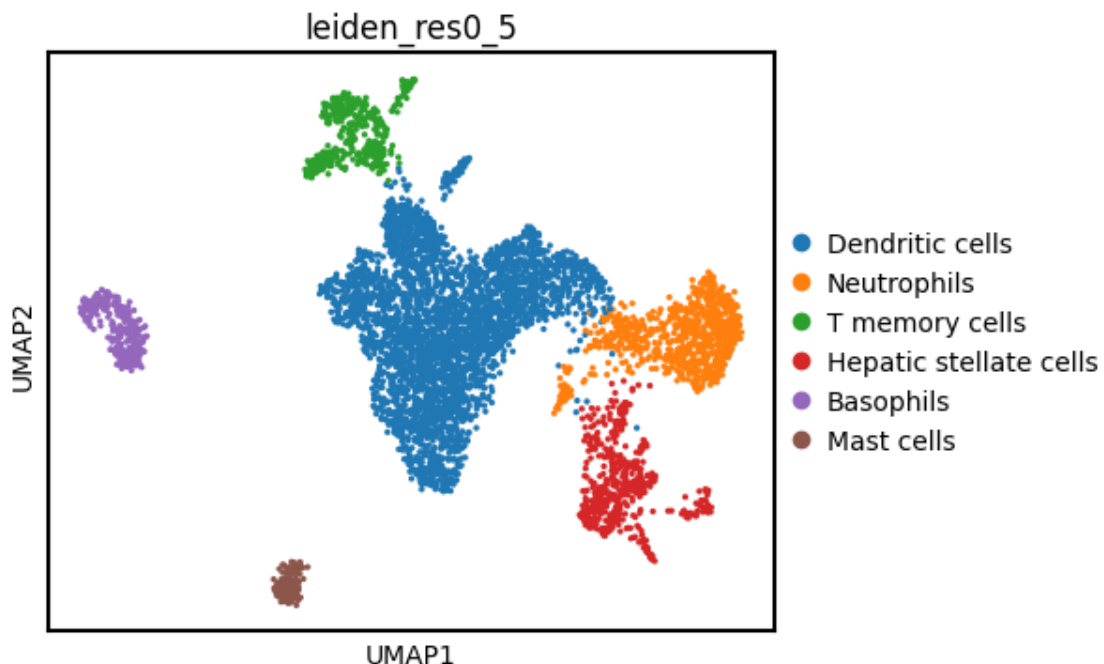
/tmp/ipython-input-2138141113.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
dict_ann = psoriasis_gene_rank[psoriasis_gene_rank["stat"] >
0].groupby("group").head(1).set_index("group")["name"].to_dict()
```

```
[ ]: {'0': 'Dendritic cells',
      '2': 'T memory cells',
      '1': 'Neutrophils',
      '3': 'Hepatic stellate cells',
      '4': 'Basophils',
      '5': 'Mast cells'}
```

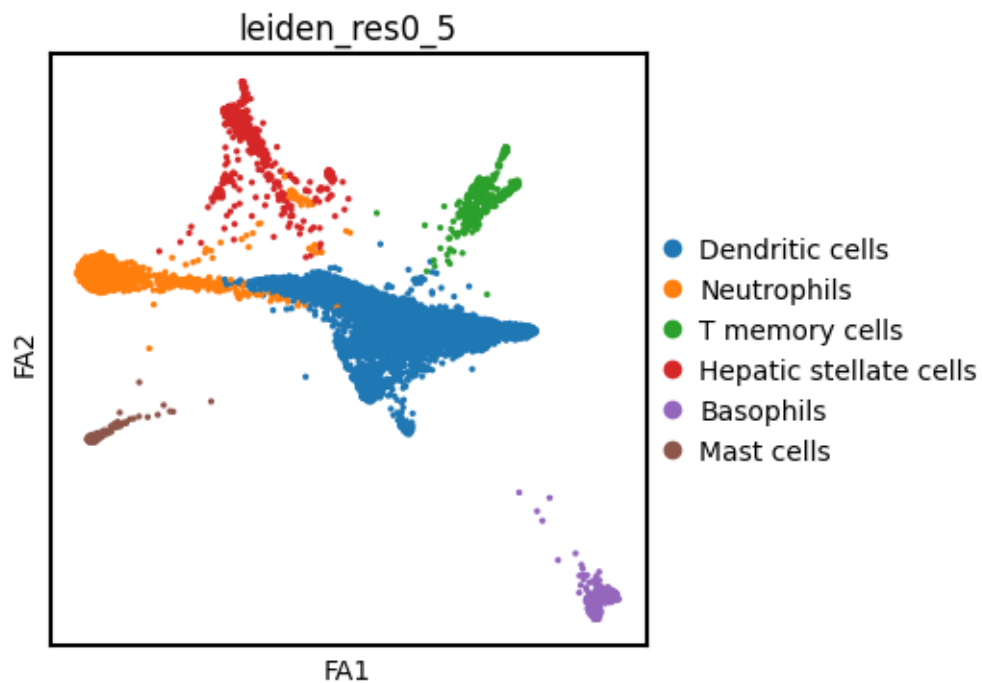
```
[ ]: psoriasis_mtx.obs["leiden_res0_5"] = psoriasis_mtx.obs["leiden_res0_5"].cat.
      ↪rename_categories(dict_ann)
```

```
[ ]: sc.pl.umap(
      adata=psoriasis_mtx,
      color=[ "leiden_res0_5"],
      ncols=1,
      )
```



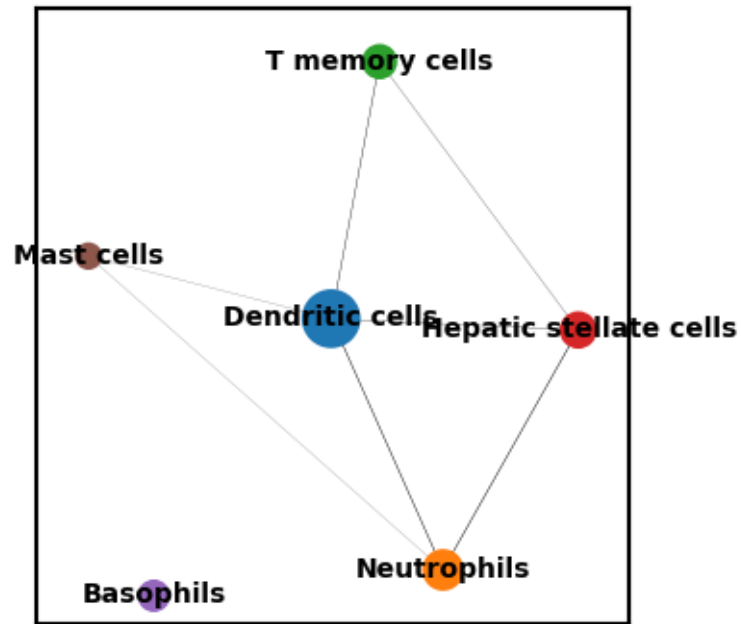
```
[ ]: #Trajectory analysis
sc.tl.draw_graph(psoriasis_mtx)
```

```
[ ]: plt.rcParams["figure.figsize"] = (4,4)
sc.pl.draw_graph(psoriasis_mtx, color='leiden_res0_5')
```



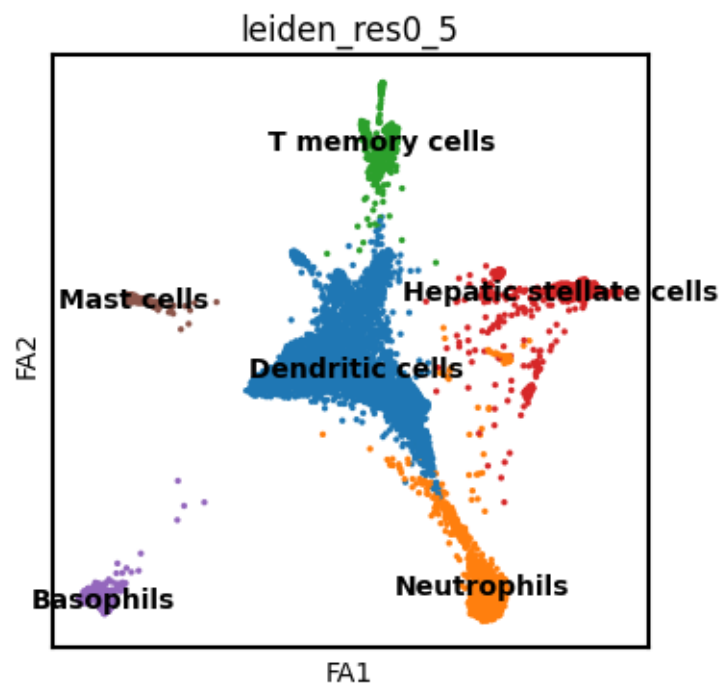
```
[ ]: sc.tl.paga(psoriasis_mtx, groups='leiden_res0_5')
```

```
[ ]: sc.pl.paga(psoriasis_mtx, color=['leiden_res0_5'])
```

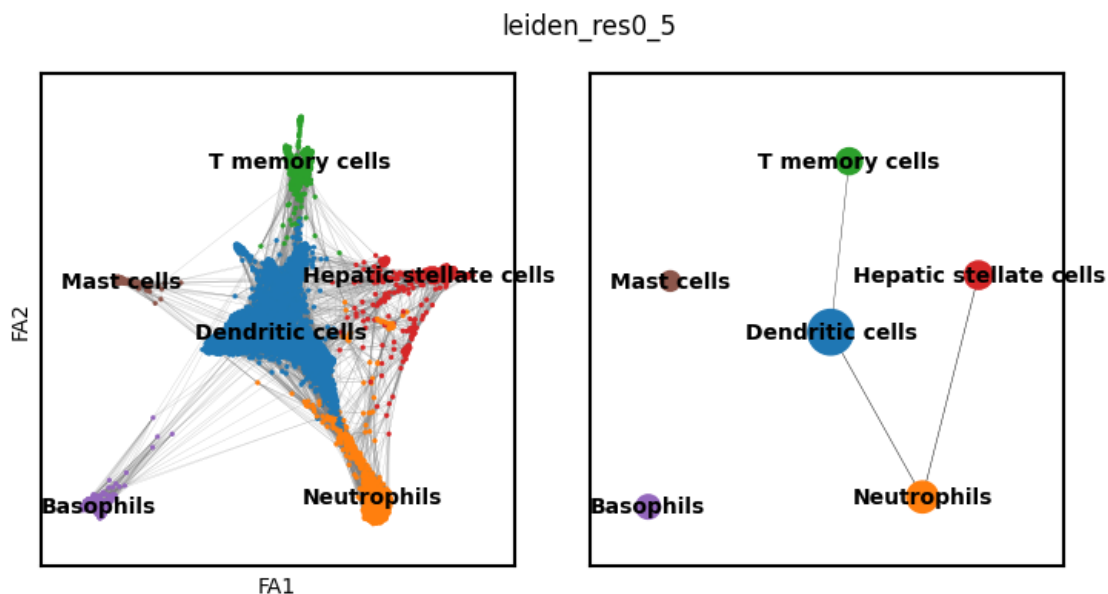


```
[ ]: sc.tl.draw_graph(psoriasis_mtx, init_pos='paga')
```

```
[ ]: sc.pl.draw_graph(psoriasis_mtx, color='leiden_res0_5', legend_loc='on data')
```




```
[ ]: sc.pl.paga_compare(psoriasis_mtx, threshold=0.03, frameon=True, edges=True)
```



```
[ ]: [<Axes: xlabel='FA1', ylabel='FA2'>, <Axes: >]
```

```
[ ]: psoriasis_mtx.uns['iroot'] = np.flatnonzero(psoriasis_mtx.obs['leiden_res0_5'] == 'T memory cells')[0]
      sc.tl.dpt(psoriasis_mtx)
```

WARNING: Trying to run `tl.dpt` without prior call of `tl.diffmap`. Falling back to `tl.diffmap` with default parameters.

```
[ ]: sc.pl.draw_graph(psoriasis_mtx, color=['dpt_pseudotime'], legend_loc='on data')
```

